

Full-length article

Renewable energy communities and the ecological transition: A game theoretic bargaining approach[☆]Leonardo Becchetti^a, Francesco Salustri^{b,c}^{*}^a University of Rome Tor Vergata, Italy^b Roma Tre University, Italy^c University College London, United Kingdom

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ABSTRACT

Renewable energy communities (RECs) are considered a key initiative for ecological transition by international institutions and policymakers. In our article, we analyze how RECs enhance cooperation in ecological transition with a game-theoretic approach. Depending on specific conditions, the interaction among RECs members can be modeled through a Prisoner's Dilemma or a game where cooperation is the only equilibrium. A REC creates incentives for cooperation through three primary revenue sources: cost savings from self-produced and consumed energy, surplus energy sales, and government incentives. We identify the combination of these factors that leads to a cooperative Nash equilibrium, where the dominant strategy for prosumers is to create a REC with passive consumers.

We further include bargaining in our model to explain how revenues are distributed between prosumers and passive consumers. Our results indicate that profit-sharing dynamics are influenced by factors such as self-consumption subsidies, energy pricing, and energy storage technology. Finally, we identify key policy parameters that influence REC viability, demonstrating that well-calibrated government incentives can significantly enhance participation and cooperation. Our study fills a gap in the literature by offering a formal strategic model that explains both the coordination mechanisms enabling successful REC formation and the bargaining processes that determine revenue distribution.

1. Introduction

The urgency of the ecological transition requires not only technological solutions but also institutional innovations that can foster cooperation among diverse energy actors. As global climate goals hinge on the rapid decarbonization of energy systems, a key challenge lies in overcoming the coordination failures that often arise in the adoption of decentralized energy supply from renewable resources. Despite increasing attention to Renewable Energy Communities (RECs) as local instruments for participatory energy transitions, there is a need for further investigation into the strategic dynamics that govern their formation and success. More specifically, most current models remain focused on technical feasibility or economic returns, without explicitly addressing the underlying game-theoretic dilemmas that characterize investment and participation decisions. This gap is particularly significant at a time when governments are crafting new incentives and

regulatory frameworks to support RECs as pillars of national climate strategies. By modeling RECs within a formal game-theoretic and bargaining framework, this study addresses the urgent need for theoretical clarity on how individual incentives, cooperative mechanisms, and policy instruments interact to determine the viability and fairness of community-led energy initiatives.

Renewable energy communities are groups of individuals, organizations, or local governments that come together to promote, develop, and implement renewable energy projects in a specific geographic area or community. Their importance is emphasized in the European Union (EU) strategy for long-term climate neutrality, where at least 75 percent of total energy demand is expected to come from renewable sources by 2050, and within this, around 16 percent of electricity generation is expected to come from collective projects (Reis et al., 2021; European Commission, 2018). The development of RECs in the European Union has recently been regulated by the EU 2018/2001/EU Directive

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^{*} Corresponding author at: Roma Tre University, Italy.

E-mail addresses: becchetti@economia.uniroma2.it (L. Becchetti), francesco.salustri@uniroma3.it (F. Salustri).

(Revised Renewable Energy Directive—RED II), but their presence also extends to many other countries. Reis et al. (2021) report that RECs generate about 11 percent of the total energy power in the United States, with a key electrification role in low-density areas.

International organizations and policymakers consider RECs to be of strategic importance for the ecological transition for several reasons. First, RECs are most often locally owned and operated, so that the benefits of clean energy production, such as cost savings and economic development, are more likely to stay within the community. Thus, the community has more control over their energy supply and can make decisions that align with their values and priorities, fostering a democratic energy transition (Hoicka et al., 2021). Second, communities can significantly contribute to increasing the adoption of renewable energy by making it more accessible and affordable for individuals and businesses. By pooling resources and sharing the costs of renewable energy projects, such as solar or wind installations, the community can reduce barriers to adoption, such as high upfront costs. Third, RECs positively contribute to public engagement and education, thereby reducing NIMBY (Not in My Back Yard) resistance to renewables. Direct participation in the benefits of ecological transition makes community members aware of its importance and potential (see also Tricarico, 2021 on this point). Fourth, RECs encourage decentralization of energy production which means increasing the level of resilience of the energy system. From this perspective, RECs, by increasing the share of prosumers (individuals who produce and consume energy locally), provide flexibility services to the electric grid, thereby reducing congestion risk in an era of progressive electrification of energy sources and an increase in the number of energy producers.

While several studies have examined the technical and economic viability of RECs, existing research lacks a formal theoretical model to understand the strategic interactions that shape their success. Our work fills this gap by introducing a game-theoretic framework to model the decision-making process of REC participants. Unlike previous research that treats the ecological transition as a social dilemma, such as the classic multiplayer Prisoner's Dilemma, we demonstrate that RECs modify the strategic environment, transforming the ecological transition into a cooperative game under specific conditions.

The paper also provides a unique perspective on the relationship between prosumers, who produce energy, and passive consumers, who only consume it, by analyzing how the bargaining power of the two actors affects revenue distribution. This aspect is crucial because fair revenue allocation determines participation incentives and, consequently, the overall stability of RECs. We also identify key policy levers, such as self-consumption subsidies and energy pricing, that influence the viability of RECs, providing policymakers with practical insights to enhance cooperation in the ecological transition. RECs create incentives for cooperation through three primary revenue sources: cost savings from self-produced energy, surplus energy sales, and government incentives. We show that these factors can lead to a cooperative Nash equilibrium, where the dominant strategy for prosumers is to establish a REC, thereby fostering collaboration in the energy transition.

We finally introduce a bargaining model that explains how surplus revenues are distributed between prosumers and passive consumers. Our results indicate that profit-sharing dynamics are influenced by factors such as self-consumption subsidies, energy pricing, and energy storage technology. We identify key policy parameters that influence the viability of REC, demonstrating that well-calibrated government incentives can significantly improve participation and cooperation. The theoretical structure of the game, bargaining model, and results are applied to a case study that illustrates how the model can be used to explain and interpret real-life experiences.

We adopt a game-theoretic framework integrated with a bargaining model because these tools are particularly well suited to address the two central challenges in the development of RECs: coordination failure and distributional conflict. Ecological transition is characterized by strategic interdependence among heterogeneous actors, where

individually rational choices may lead to collectively suboptimal outcomes. In this context, a game-theoretic model captures how incentive structures—such as subsidies and surplus pricing—can reshape strategic payoffs and enable cooperative equilibria. Moreover, a REC involves a negotiation over surplus distribution between prosumers, who bear the investment cost, and passive consumers, who participate without incurring such costs. Since this process involves asymmetric contributions and outside options, a bargaining model is methodologically necessary to analyze how revenues can be shared in a way that sustains participation and avoids conflict.

Previous research has investigated how community members should optimally allocate costs and profits of the shared infrastructure needed for a REC (Moncecchi et al., 2020; Gomes and Vale, 2024). Our analysis, however, not only focuses on how energy and their revenues should be shared, but also considers the choice of joining a REC from a utility-maximizing consumer and how existing prosumers may make their REC more attractive to them. None of the previous research contributions have explicitly examined the strategic relationship between prosumers and passive consumers using a game-theoretic model. By filling this research gap, our study advances understanding of how RECs can be structured to maximize participation, efficiency, and fairness, ultimately contributing to a more effective and inclusive ecological transition.

2. Literature review

A recent review of the academic literature on energy communities in Europe finds that most studies focus on renewable energy produced from photovoltaic energy (Gianaroli et al., 2024). In addition to economic incentives, several key drivers and challenges influence the success of a REC. Economic savings on energy costs represent a key factor, but consumers also pay attention to the social dimension (Caferri et al., 2024).

Previous literature contributions show that a complex interaction among financial incentives and ownership, social acceptance, and the policy framework determines the feasibility of a REC. In Europe, successful RECs often rely on strong financial participation from local members, well-structured cooperative business models, and stable policy environments that provide investment security and access to the priority network (Bauwens et al., 2016). Based on these insights, Oteman et al. (2014) find significant differences in the effectiveness of RECs in Denmark and Germany, primarily due to the influence of national regulatory frameworks. Denmark has developed a highly successful REC model through long-term policy support, easy access to the grid, and mandatory local ownership requirements, ensuring that communities directly benefit from renewable energy production. In contrast, Germany experienced a boom-and-bust cycle, where substantial feed-in tariffs initially propelled REC growth, but later policy reversals (such as 2014 subsidy cuts) introduced market uncertainties that hindered expansion. The Netherlands, on the other hand, exhibited slower REC growth due to bureaucratic hurdles, limited financial incentives, and the dominance of centralized energy companies.

In addition to policy stability, Hicks and Ison (2018) emphasizes the importance of financial mechanisms, noting that access to cooperative financing models, subsidies, and feed-in tariffs significantly influences REC participation rates. More recently, Janev et al. (2025) have examined how digitalization and smart grid integration can enhance REC efficiency, making them more attractive to investors.

As community-driven initiatives, RECs are highly reliant on social cohesion and trust. Bauwens et al. (2016) demonstrates that social capital, as defined by trust, local engagement, and participatory governance, plays a crucial role in determining the long-term success of REC's projects. Participation can also be enhanced or prevented by ownership and governance choices. In a study conducted in France in 2024, REC members declared that they valued the opportunity to delegate their voting rights within the community to a trustee,

but disliked a capital-based voting rule and co-investments by other firms (Guetlein and Schleich, 2024). Devine-Wright et al. (2017) emphasizes the importance of collective action and social innovation, arguing that RECs thrive when community members actively participate in decision-making processes. Similarly, Haf and Parkhill (2017) find that projects that prioritize fairness, inclusivity, and shared economic benefits tend to be more resilient, particularly in overcoming opposition from established energy providers.

Despite their advantages, many RECs face financial barriers. As noted in Bauwens et al. (2016), the high upfront costs of renewable energy infrastructure and the reluctance of financial institutions to fund community-led projects can pose major obstacles. Market competition is another complication; large utility companies often resist the expansion of RECs, lobbying for regulatory frameworks that maintain their market dominance.

A growing body of research has explored the role of RECs in improving energy resilience, particularly in the face of climate change and concerns about energy security. Zucaro and Agostinho (2025) argue that RECs can increase urban sustainability by promoting local energy autonomy, reducing dependence on centralized power systems, and improving resilience against grid failures. By decentralizing energy production, RECs contribute to a more distributed and adaptive energy system, which is particularly valuable in regions prone to climate-related disruptions. Furthermore, advances in smart grid technologies and decentralized energy trading have opened new opportunities for RECs. Debnath et al. (2024) discusses the integration of blockchain-based energy markets, which allow communities to participate in peer-to-peer energy exchanges, optimizing resource distribution and financial returns. Recent literature continues to emphasize the multifaceted role of cooperation and Renewable Energy Communities (RECs) in supporting decentralized energy transitions. Liu et al. (2025) develops a regional cooperation model using game theory to assess the effectiveness of green finance mechanisms and local governance in China's low-carbon transition, suggesting that interregional REC cooperation can be optimized with proper policy support and credit allocation schemes. Advances in smart grid integration also contribute to improved REC outcomes. Koirala et al. (2016) highlights how integrated community energy systems can improve both technical efficiency and fairness in revenue allocation. How RECs are organized and embedded within the political context matters, too. Cooperative structures are more successful when linked to broader environmental citizenship initiatives like net-zero-energy buildings (Dall'Orsoletta et al., 2022; Minelli et al., 2024), suggesting that social innovation and energy democracy remain central to REC proliferation. The role of investors is also relevant for RECs. Cielo et al. (2021) analyzes the financial and environmental performance of different RECs' business models in Italy and finds positive performances under both direct community investment and investment by a technological partner that acquires and manages the assets. In addition to the financial performance, Hanke et al. (2021) documents and investigate the capacity of a sample of 71 RECs to include underrepresented and vulnerable groups and mitigate energy poverty as a particular form of energy justice.

3. Research hypothesis and the gap bridged in the existing literature

3.1. Research questions

The objective of our work is to answer the following three research questions: Can renewable energy communities contribute to solving the investment coordination failure in ecological transition? What factors can favor or hinder this outcome? What factors affect the bargaining power of prosumers and passive consumers when sharing revenues?

While several papers have addressed the general question of what fosters or hinders the creation of RECs, none of these studies is based on a game theoretical model addressing the social dilemma of ecological

transition in the case of RECs and, more importantly, no research focuses on the strategic relationship between prosumers and active consumers. Filling this gap is of paramount importance, as profit sharing between RECs participants (i.e., prosumers and passive consumers) is what ensures citizens participation in RECs and prevents resistance from NIMBYs to the ecological transition.

The game theoretical approach applied to ecological transition provides a relevant theoretical background as it allows for modeling coordination among a large number of players, which is required to tackle climate change. A growing body of literature models climate action as a social dilemma, where individual rational choices lead to collectively suboptimal outcomes unless cooperation is incentivized.

One of the foundational contributions in this area comes from Decanio and Fremstad (2013), who examine the strategic structure of climate negotiations. The authors argue that climate negotiations can be modeled as a Prisoner's Dilemma (PD), where nations face incentives to free-ride on others' mitigation efforts, rather than a simple coordination game. Because cooperation often fails, international diplomacy should emphasize the urgency of climate risks, shifting the game towards a coordination framework that fosters cooperative equilibria.

Building on this approach, Becchetti and Salustri (2019) analyze environmentally responsible choices through a social dilemma framework, exploring conditions under which responsible actions can become a Nash Equilibrium (NE). Their findings highlight the role of costs and incentives in making mutually responsible choices an equilibrium strategy.

More recently, ecological interactions have been further explored by Carrozzo Magli and Manfredi (2022), who emphasize how short-term preferences and the intertemporal impact of current choices on future payoffs make policy decisions difficult to achieve. Their analysis remarks on the need for institutional mechanisms that counteract the myopic behavior of individuals and promote sustainable strategies.

Beyond environmental choices, Alvarez et al. (2019) apply cooperative game theory to specific ecological threats, modeling river flood risk management, showing how structured cooperation can enhance both prevention and mitigation efforts. Climate actions involve not only coordination among participants and incentives to cooperate, but also perception about the risks associated with climate change and expectations on other players' behavior. In this direction, Becchetti et al. (2025) expands the social dilemma framework and examines how people's beliefs about climate threats and expectations of others' actions shape their willingness to act ecologically. Empirical findings suggest that both urgency and awareness of others' actions have a positive impact on ecological commitment. The authors interpret this as evidence that intrinsic motivations are socially influenced, with social capital moderating the effect.

In general, these studies highlight the relevance of game theory in modeling ecological transition as a social dilemma, providing crucial insights on incentives, perceptions, and institutional conditions that can foster global cooperation in the fight against climate change.

3.2. Bridging the research gap

Within this framework, our paper fills a significant research gap in the literature on renewable energy communities (RECs) by introducing a game-theoretic approach to model the ecological transition in this specific context. Existing studies primarily treat the ecological transition as a Prisoner's Dilemma (PD), where free-riding incentives create a coordination failure. Our paper frames this perspective by showing how RECs can modify the strategic environment, transforming ecological transition from a PD into a Stag Hunt Game, or even an original cooperative game where collective action becomes the dominant strategy under specific conditions.

Unlike previous research that examines RECs primarily from economic feasibility or technical perspectives (Moncecchi et al., 2020;

Gomes and Vale, 2024; Velkovski et al., 2024), we provide a formal theoretical framework explaining how the structure of revenues (through energy savings, surplus sales, and government subsidies) affects strategic cooperation. We also identify key conditions that determine whether RECs succeed or fail due to coordination dynamics, offering a new dimension to the study of RECs as collective decision-making mechanisms. More importantly, we address a fundamental gap in the literature by modeling the bargaining process between prosumers and passive consumers. Although previous work has analyzed the policies and governance aspects of RECs (Hoicka et al., 2021; Reis et al., 2021), none have explicitly studied the strategic relationship between these two groups, despite their interaction being crucial to ensure fair profit sharing, long-term stability, and citizen participation. By formalizing this bargaining process, we highlight the factors that shape bargaining power, illustrating how an unbalanced division of revenues could undermine participation and potentially trigger NIMBY resistance.

Our contribution goes beyond theoretical modeling by illustrating a case study showing how the model fits the reality (in Section 7) and providing strategic insights for policymakers. Unlike previous studies that focus on economic incentives in isolation, our analysis demonstrates how self-consumption subsidies, energy pricing, and government incentives interact to influence the emergence of cooperative equilibria. This perspective offers critical policy implications, particularly regarding how to establish financial incentives at levels that maximize cooperation while controlling public expenditure. Furthermore, our findings underscore the importance of balancing bargaining power between prosumers and passive consumers, ensuring that RECs are not only economically viable but also socially inclusive. By applying game theory to RECs, we enhance our understanding of how these communities can overcome coordination failures in investment for ecological transition, thereby filling a crucial gap in the literature and providing a novel framework for analyzing bargaining dynamics that are essential for ensuring equitable participation and long-term sustainability.

4. Using game theory to assess the role of Renewable Energy Communities in ecological transition

The economic literature on ecological transition highlights how many choices related to it can be analyzed in the context of multiplayer social dilemmas (Becchetti and Salustri, 2019). Buying a full-electric or plug-in car instead of the non-electric substitute currently entails extra costs, even though it is expected that economies of scale will highly likely reduce the cost differential over time, and the total cost of ownership differential between the two car types will be progressively lower for longer car possession and distance traveled, since operating costs tend to be lower for electric vis-à-vis petrol cars. The characteristics of the Prisoner's Dilemma in this case are such that if all players chose ecologically, everyone would be better off in terms of lower emissions and reduced contribution to the threat of climate change. However, when the cost differential between the two car types (even after considering the total cost of ownership) applies, the dominant strategy for a player without other-regarding preferences and solely focused on their own monetary payoff is to buy the non-electric alternative. We therefore fall into a classic multiplayer Prisoner's Dilemma where the Nash equilibrium outcome is the crossing of players' dominant strategies and is paradoxically Pareto dominated, that is, every player would be better off under the alternative 'ecological cooperative' choice where everyone buys the electric car (see Table 1 for a typical Prisoner's Dilemma payoff structure).

A similar reasoning can be applied when players have to choose between house retrofit against the alternative of doing nothing where, in absence of substantial environmental subsidies, the benefits of an investment in energy efficiency (lower heating costs due to better insulation, increase in the value of the building when it achieves a

Table 1
The Prisoner's Dilemma structure.

		Consumer 2	
		Ecological	Non-ecological
Consumer 1	Ecological	(3, 3)	(0, 5)
	Non-ecological	(5, 0)	(1, 1)

Table 2
The Stag Hunt structure.

		Consumer 2	
		Ecological	Non-ecological
Consumer 1	Ecological	(3, 3)	(0, 1)
	Non-ecological	(1, 0)	(1, 1)

higher energy class) less than compensates the investment cost at least in the short run.

To simplify the analysis in the case of two players, the valuable difference from the two previous examples in the case of RECs is that they can easily become a Stag Hunt game instead of a prisoner's dilemma, thus increasing the probability of a cooperative outcome. The Stag Hunt game's difference lies in its payoff structure, where there are no dominant strategies. In the presence of a cooperative choice by one player, the best choice for the other player is to cooperate rather than defect. Thus, the game has no dominant strategies and two Nash equilibria (cooperate, cooperate) and (defect, defect), where the first Pareto dominates the other (see Table 2 for a typical Stag Hunt game payoff structure).

More in detail, and as we will explain in the model presented in the next section, in the RECs game we have two different players, prosumers and passive consumers, where the first brings the investment in solar panels (or in any case in renewable energies such as wind turbines) on a productive site (a roof in case of solar panels, an area where a wind turbine can be installed in the second case) to the REC, while the second is just a passive energy consumer. The characteristic of the REC model is that, if the prosumer decides to invest, the best course of action for the passive consumer is to participate in the REC (cooperate) rather than refusing to participate (defect). This result is different from what happens in the decision to buy an electric car or to participate in a house retrofit investment where, given the Prisoner's Dilemma framework, the dominant strategy of a self-interested player is to defect and therefore defecting is the optimal choice even when the other player cooperates (e.g., buying the electric car or contributing to the house retrofit investment).

The reason for this difference is embedded in the three economic benefits of the REC: (i) saving the cost of energy that is self-produced and consumed and not purchased on the market; (ii) selling the energy surplus not consumed locally to the energy grid manager; (iii) earning the self-consumption subsidy provided by the government for the flexible electric grid service provided by local consumption of the energy produced. Given these three benefits, the prosumer's decision to cooperate (i.e., invest in a production site and allocate the energy produced towards the REC) automatically makes it convenient for the passive consumer to cooperate (i.e., enter the REC). In contrast, the passive consumer's decision not to enter the REC would lead to a lower payoff for both players than under the cooperation alternative. However, in this case, achieving the Pareto superior equilibrium requires more coordination than a simultaneous one-shot game, such as a preliminary binding contract among parties.

The cooperative solution is even easier to achieve when the prosumer's utility function is such that it is convenient for her to invest in the productive plant even if passive consumers do not participate in the venture (Table 3 shows how in this case investing in the REC is the dominant strategy of the prosumer).

When this is the case, we exit the Stag Hunt game region, and the cooperative equilibrium is more easily attainable. In this framework,

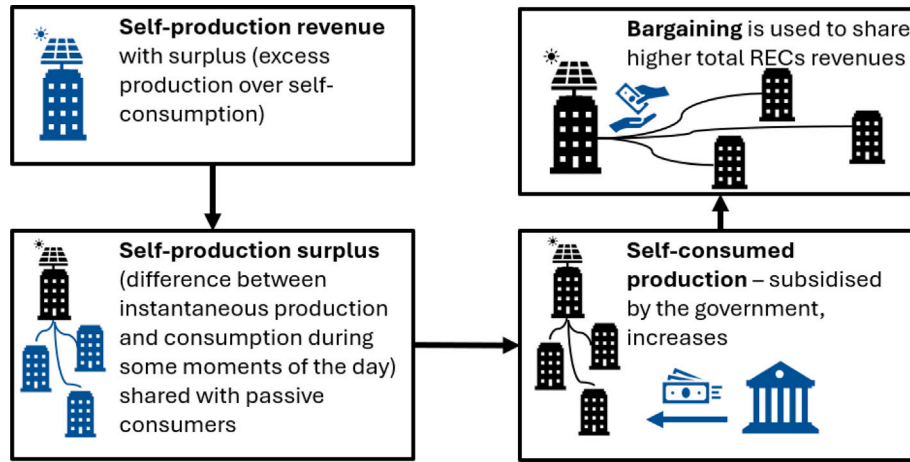


Fig. 1. Why renewable energy can be a cooperative game.
Source: Author's elaboration.

Table 3
The REC game structure.

		Passive consumer	
		Ecological	Non-ecological
Prosumer	Ecological	(3, 3)	(2, 1)
	Non-ecological	(1, 0)	(1, 1)

the choice of both players to participate in the REC is the unique Nash equilibrium of the simultaneous game. If we were to make the game sequential, the prosumer would make the first step of investing in the production site, as investing becomes their dominant strategy. After this move, the passive consumer would find it optimal to participate in the REC. When this is the case, the strategic problem of the REC, therefore, is not in the choice between creating or not creating the REC, but in how the REC's profits should be distributed between prosumers and passive consumers, considering the distinct nature of the two types of players. More specifically, a prosumer pays the investment cost and therefore needs to cover costs with REC revenues, whereas the problem does not occur for passive consumers, who can, however, incur a fixed switching cost for entering the REC.

Fig. 1 represents why REC can be modeled as a cooperative game, where prosumers and passive consumers can increase the share of self-consumed production which the government remunerates.

In the section that follows, we outline the problem by describing characteristics of the REC profit function and then outline a bargaining solution over the REC profits between the two types of players.

5. The model

The model presented in this section makes use of the variables described in Table 4.

A Renewable Energy Community (REC) is a community of N households, of which M own the energy production plants (*active prosumers*, henceforth APs) and $N - M$ do not (*passive consumers*, henceforth PCs).

The choice of households When a REC is established by prosumers, other households without energy plants can choose among three options: (1) join a REC as an AP by investing in an energy plant; (2) join a REC as a PC; (3) remain in the national market and buy all energy from the market as a *standard consumer* (henceforth, SC).

Production Each AP i produces the energy output y_i using her plant, and this energy can be either consumed by REC members (i.e., APs and PCs) or sold on the market; the energy production capacity of the whole REC is $Y \equiv Y^M := \sum_{i=1}^M y_i$. APs incur investment costs, I , required to

plan, install, and operate the energy production plant, and are financed by external sources (e.g., bank financing) at an interest rate, r . The energy prosumed (i.e., produced and simultaneously consumed by REC members, APs and PCs) is a share α of the total REC production Y and is rewarded at price p_k by the government for the electric grid flexibility service implicitly provided by production that is consumed locally and does not increase traffic on the electric grid. The rest of the energy $(1 - \alpha)$ is sold in the market at the price p_R .

A few further elements can be embedded into this model. First, energy production is not constant throughout the day, with central, sunlight hours producing high levels of energy while nighttime, dark hours produce low or no amount of energy. In this case, we may assume two periods, a light period $t = 1$, with production Y_1 , and a dark period $t = 2$ with production Y_2 , where $Y_1 > Y_2$. A further distinction can be made between energy prosumed and energy accumulated (i.e., produced by REC members and stocked for delayed consumption). This energy requires accumulation technologies and is not subsidized by the government (this is how the subsidy was designed in Italy, see the [CER Decree Law, 2023](#)). In this case, we assume that $\alpha = \alpha_1 + \alpha_2$, with α_1 being the share of the energy prosumed and α_2 the share of energy accumulated.

Consumption. Each household consumes c_i energy, with $c_i \in [0, +\infty)$. The total energy consumed by REC members is $C \equiv C^N := \sum_{i=1}^N c_i$. This energy can be prosumed, accumulated, or purchased by the market at a price p_G . Similarly to production, the energy consumption is not constant over the day.

Preferences. Households members of a REC have the following three aggregate sources of benefits: (i) the *opportunity benefit* – they consume a share of the energy produced by the REC plants without buying it from the external market at price p_G ; (ii) the *direct benefit* – they sell the surplus of the energy produced to the domestic electrical grid manager at price p_R ; (iii) the *subsidy benefit* – they enjoy the payment provided by the government as a subsidy, which equals a given price p_k times the prosumed energy, i.e., $p_k \alpha Y^M$. Each REC member may have ordered preferences over these three aggregate sources of benefits. Household members choose these preferences when they establish a REC, and in the next section, we analyze the bargaining process of such a choice.

Fig. 2 illustrates how the energy produced and consumed in a REC works over a day. During the most intense period of consumption, community members (APs and PCs) consume both the energy they produce (blue area) and the energy purchased from the electricity supplier (red area on the left). This result is because we assume the capacity of a REC may not be enough to satisfy the demand of the REC members at that moment of the day. Then, when the electricity demand falls, community members can consume the energy produced

Table 4

Variable legend.

Variable	Description
AP	Active prosumer
PC	Passive consumer
y_i	Energy produced by AP i 's CER plants
Y^M	Total energy produced by a CER with M APs.
α	Share of prosumed energy (produced by M APs and consumed by all N members)
c_i	Energy consumed by i 's member of the CER (regardless of how it is produced)
C_i^N	Total energy consumed by N CER members
γ	Share of the prosumed energy αY which is consumed by APs.
R_α	Total revenues for the REC that depend on the share of prosumed energy α .
p_k	Government subsidy for the energy prosumed
p_G	Energy price in the external market
p_R	Energy price between REC and the energy supplier

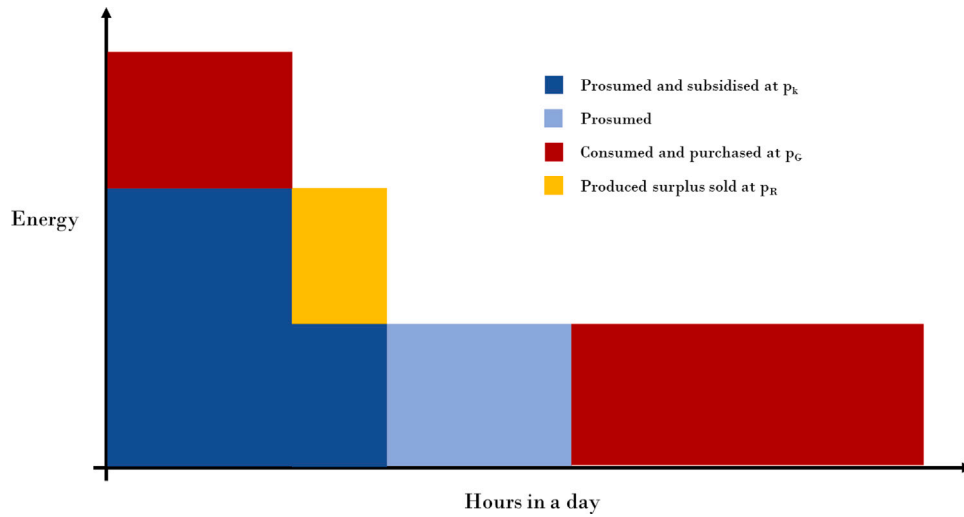


Fig. 2. Energy consumption, production, and sale patterns in the REC.
Source: Author's elaboration.

by the REC and, at the same time, sell the energy surplus (yellow area). Later, when the REC plants no longer produce energy, community members can first consume the energy they have previously produced and accumulated (light blue area), and then they can buy the energy from the market (red area on the right).

Revenues Based on the definition of the aggregate sources listed above, the total REC revenues will therefore be equal to

$$R_\alpha = p_k \alpha Y + p_R (1 - \alpha) Y \quad (1)$$

with $0 < \alpha < 1$. In other words, revenues are made by the prosumed energy subsidies at p_k and the energy surplus sold at the local supplier at p_R , and they depend on the share of energy prosumed α .

Prosumer's utility. Among all REC members, APs prosume a share γ of the αY energy prosumed by the REC, as opposed to that consumed by PPs (i.e., $1 - \gamma$). Given the energy consumed by each household c_i and assuming each AP consumes the same share of prosumed energy, i.e., $\frac{1}{M} \gamma$, the utility of AP i is equal to

$$U_i^{AP} = \frac{\gamma}{M} R_\alpha - p_G (c_i - \frac{\gamma}{M} \alpha Y) - I(1 + r) \quad (2)$$

where $(c_i - \frac{\gamma}{M} \alpha Y)$ is the energy consumed by the prosumer that is not produced by them and therefore purchased at the national market, $I(1 + r)$ has been defined above as the discounted cost of investment in energy plants, and $0 < \gamma_i < 1$. We also assume $c_i \geq \frac{\gamma}{M} \alpha Y$, that is, the prosumers do not waste energy or, in other words, the energy they produce and allocate for self-consumption is all self-consumed. This

pattern must be maintained throughout the day, as we observe that there may be some hours during the day when c_i exceeds y_i for APs.

Note that (2) holds under the assumption that the REC has no accumulation capacity. If REC members have an accumulation capacity, we should unpack the produced energy component into prosumed and accumulated energy. In this case, the utility function of the prosumer becomes

$$\hat{U}_i^{AP} = \frac{\gamma}{M} R_\alpha - \frac{\gamma}{M} \alpha_2 Y p_K - p_G (c_i - \frac{\gamma}{M} (\alpha_1 + \alpha_2) Y) - (I + I_c)(1 + r) \quad (2^A)$$

where we subtract $\gamma_i \alpha_2 p_K Y$ since we have assumed the stocked energy consumed later is not subsidized, and we add the cost of the accumulation technology, I_c .

Passive consumer's utility. In the absence of accumulation technology, assuming each PC consumes the same share of prosumed energy, i.e., $\frac{(1-\gamma)}{N-M}$, the utility of the i th PC is equal to

$$U_i^{PC} = \frac{1-\gamma}{N-M} R_\alpha - p_G (c_i - \frac{(1-\gamma)}{N-M} \alpha Y) \quad (3)$$

Unlike prosumers, passive consumers do not incur investment costs.

Standard consumer's (SC) utility. The i th SC purchases energy c_i from the national market at price p_G and therefore her utility will be

$$U_i^{SC} = -p_G c_i \quad (4)$$

The optimal choice for the passive consumer. The passive consumer can choose to leave the REC and become a SC, to remain in the REC as a PC, or to improve her role in the REC, invest in an energy plant, and become an AP.

More specifically, the PC will remain in the REC as PC as long as her utility is higher than the utility of SC, that is,

$$\underbrace{\frac{(1-\gamma)}{N-1-M} R_\alpha - p_G(c_i - \frac{(1-\gamma)}{N-1-M} \alpha Y)}_{U_i^{PC}} \geq \underbrace{-p_G c_i}_{U_i^{SC}} \quad (5)$$

which can be written in terms of the market price of the energy as

$$p_G \alpha Y \geq -R_\alpha \quad (6)$$

If the PC wants to invest and become AP, they will do so if and only if

$$\begin{aligned} & \underbrace{\frac{\gamma}{M+1} R_\alpha - p_G(c_i - \frac{\gamma}{M+1} \alpha Y) - I(1+r)}_{U_i^{AP}} \\ & \geq \underbrace{\frac{1-\gamma}{N-M} R_\alpha - p_G(c_i - \frac{(1-\gamma)}{N-M} \alpha Y)}_{U_i^{PC}} \end{aligned} \quad (7)$$

that can be written as

$$\frac{\gamma}{M+1} R_\alpha - p_G(\frac{\gamma}{M+1} \alpha Y) - I(1+r) \geq \frac{1-\gamma}{N-M} R_\alpha - p_G(\frac{(1-\gamma)}{N-M} \alpha Y) \quad (8)$$

that is,

$$I(1+r) \leq (R_\alpha - p_G \alpha Y) \left(\frac{\gamma}{M+1} - \frac{1-\gamma}{N-M} \right). \quad (9)$$

Eq. (9) requires the REC to have two main, necessary conditions for the PC to become an AP: first, the revenues from the REC must be greater than the cost of the equivalent amount of energy in the energy market; second, the amount APs take from the prosumed energy must be greater than the amount that PCs take from the prosumed energy.

The role of model parameters on making a Prisoner's Dilemma a REC game. When all PCs choose to join the REC as APs, we have $\gamma = 1$ and $M = N$. Then, (9) becomes $I(1+r) \leq (R_\alpha - p_G \alpha Y) \frac{1}{N}$, which says that the investment must be repaid with the profits (i.e., revenues minus costs). In this case, the game falls into in the REC game and APs find convenient to start the REC even in absence of passive consumers; if $(R_\alpha - p_G \alpha Y) \frac{1}{N} \leq I(1+r) \leq (R_\alpha - p_G \alpha Y) \left(\frac{\gamma}{M+1} - \frac{1-\gamma}{N-M} \right)$, the game falls into the Stag Hunt game and the creation of the REC requires coordination and a previous agreement with passive consumers.

Proposition 1 (*The REC Marginal Benefit of Passive Consumers*). *If $p_k > p_R$ and $\alpha < 1$, then any additional passive consumer increases the share of REC energy consumed by members and therefore REC total revenues.*

This is a trivial property of the system of REC incentives since, with $p_k > p_R$, REC revenues are maximized if $\alpha = 1$ and any configuration with $\alpha < 1$ is inefficient. The implication is that prosumers will campaign to find the highest possible number of passive consumers. From a game-theoretical perspective, this means that the prosumer's payoffs increase if the passive consumer cooperates, i.e., she becomes a REC member. The number of passive consumers is a necessary but not sufficient condition to increase the prosumer's payoff. It is also required that consumption patterns of passive consumers are not perfectly correlated with those of existing members, and thus, there is a production surplus that can satisfy the demand of new passive consumers. This result may have a practical policy implication. Households usually consume more energy in the evening, while schools, offices, or firms consume more energy in the morning. Therefore, optimally matching different types of consumers may increase the payoffs of REC members and maximize the prosumed energy.

The fair sharing rule. The fair sharing rule for surplus sales and self-consumption revenues among REC members requires the definition of optimal values for γ such that $U_i^{PC} = U_i^{AP}$ for each $i = 1, \dots, N$ and $\gamma > 0$.

Table 5

The REC game with bargaining payoffs.

		Passive consumer	
		Ecological	Non-ecological
Active prosumer	Ecological	(U_i^{AP}, U_i^{PC})	$(\bar{U}_i^{AP}, -p_G c_i)$
	Non-ecological	$(-p_G c_i, 0)$	$(-p_G c_i, -p_G c_i)$

If we choose ex ante, a deterministically equal-sharing rule for energy consumption among members, the problem reduces to finding γ such that $U_i^{PC} = U_i^{AP}$ for each i .

$$\gamma = \frac{I(1+r)(N-M)M}{N(R_\alpha - p_G \alpha Y)} + \frac{M}{N} \quad (10)$$

That is, the prosumer gets all the revenues up to the moment at which its investment cost is covered.

6. The model with bargaining

When both APs and PCs decide to set up a REC, it is likely that they will bargain to maximize their share of revenues. To model this feature, we adopt a bargaining model where the bargaining power has an exogenous component driven by idiosyncratic bargaining abilities and an endogenous component that depends on the value of players' outside options. Intuitively, a higher value of the gain from the agreement makes the player's bargaining power weaker. A higher value of the gain increases the disutility associated with failing to reach an agreement. On the contrary, ceteris paribus, an increase in her outside option will reduce their gain from agreement and reinforce their bargaining power (Fig. 3).

6.1. Bargaining between two prosumers

For the sake of simplicity, we assume there is one AP which has invested I to install an energy plant and produce electric energy, and another household which can choose between becoming a PC in the REC or remaining a standard energy consumer outside the REC.

More formally, if $\alpha_{(AP)}$ and $\alpha_{(AP,PC)}$ denote the share of prosumed energy with one AP, one AP and one PC, respectively, and $\alpha_{(AP,PC)} > \alpha_{(AP)}$, the AP has the following outside option

$$\bar{U}_i^{AP} = \gamma R_{\alpha_{(AP)}} - p_G(c_i - \gamma \alpha_1 Y) - I(1+r) \quad (11)$$

vis-à-vis the utility in case the other household chooses to join the REC as PP, which writes

$$U_i^{AP} = \gamma R_{\alpha_{(AP,PC)}} - p_G(c_i - \gamma \alpha_{(AP,PC)} Y) - I(1+r) \quad (12)$$

and we have that $U_i^{AP}(\gamma) > \bar{U}_i^{AP}(\gamma)$ since $\alpha_2 > \alpha_1$.

The outside option of the PC, on the other hand, is:

$$\bar{U}_i^{PC} = -p_G c_j \quad (13)$$

vis-à-vis the utility in case the PP chooses to join the REC as PP, which writes

$$U_i^{PC} = \gamma R_{\alpha_{(AP,PC)}} - p_G(c_i - \gamma \alpha_{(AP,PC)} Y) \quad (14)$$

Table 5 incorporates the bargaining payoffs into the REC game.

The bargaining share maximand is, therefore:

$$\Omega = (U_i^{AP} - \bar{U}_i^{AP})^\beta (U_i^{PC} - \bar{U}_i^{PC}) \quad (15)$$

for $i = 1, 2$, and the AP equilibrium share, if we consider gains from agreement in terms of the difference between the utility with and without agreement for each player, is equal to:

$$\gamma_j^* = \arg \max_{R_{\alpha_i}} (U_i^{AP} - \bar{U}_i^{AP})^\beta (U_i^{PC} - \bar{U}_i^{PC}) \quad (16)$$

with $\beta = 1$.

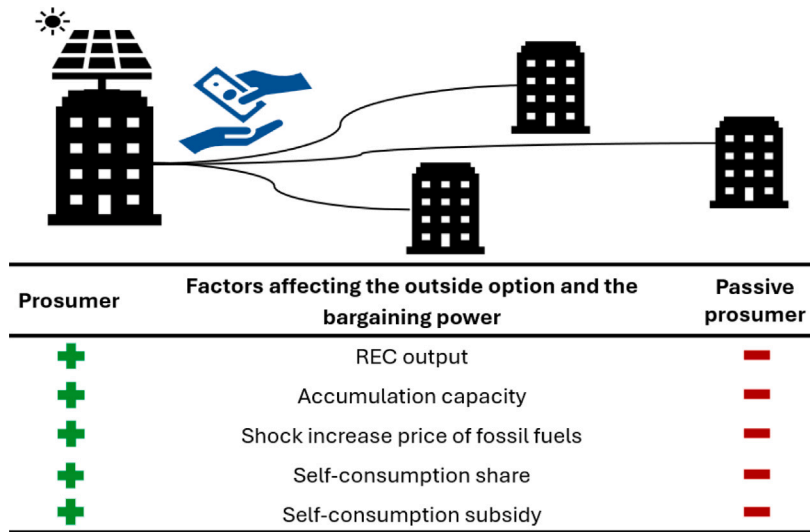


Fig. 3. Factors affecting how prosumer and passive consumers share their revenues.
Source: Author's elaboration.

This is equivalent to

$$\gamma_j^* = \arg \max_{R_{\alpha_i}} \gamma^2 \left(R_{\alpha_{AP}, PC} (R_{\alpha_{AP}, PC} - R_{\alpha_{AP}}) + p_G \alpha_2 Y (R_{\alpha_{AP}, PC} - R_{\alpha_{AP}}) + p_G Y (\alpha_{AP, PC} - \alpha_{AP}) + p_G^2 Y^2 \alpha_2 (\alpha_{AP, PC} - \alpha_{AP}) \right) \quad (17)$$

6.2. Comparative statics at equilibrium

In order to solve the bargaining problem, consider that the first-order condition is:

$$\frac{\partial \log \Omega}{\partial \gamma_j} = \frac{\beta}{U_i - \bar{U}_i} \frac{\partial (U_i - \bar{U}_i)}{\partial \gamma_j} + \frac{1}{U_j - \bar{U}_j} \frac{\partial (U_j - \bar{U}_j)}{\partial \gamma_j} = 0 \quad (18)$$

where

$$\frac{\partial (U_i - \bar{U}_i)}{\partial \gamma_j} = ((\alpha p_k Y + (1 - \alpha) p_R Y) + 2 p_G \alpha Y) > 0$$

That is, higher prosumer bargaining power creates a marginal benefit of a higher self-consumption share and a higher share of the surplus of energy produced by the REC and sold on the market. Conversely,

$$\frac{\partial (U_j - \bar{U}_j)}{\partial \gamma_j} = -((\alpha p_k Y + (1 - \alpha) p_R Y) + 2 p_G \alpha Y) < 0$$

Hence,

$$\frac{\partial (U_i - \bar{U}_i)}{\partial \gamma_j} = - \frac{\partial (U_j - \bar{U}_j)}{\partial \gamma_j}$$

Since what adds to the prosumer share is subtracted from the share of the passive consumer. As a consequence, $\frac{\partial \log \Omega}{\partial \gamma_j} = 0$ implies:

$$\begin{aligned} & \beta((\alpha p_k Y + (1 - \alpha) p_R Y) + 2 p_G \alpha Y)(U_i - \bar{U}_i) \\ & - (\alpha p_k Y + (1 - \alpha) p_R Y - 2 p_G \alpha Y)((1 - \gamma_i)(\alpha p_k Y + (1 - \alpha) p_R Y) + \\ & p_G(c_j - (1 - \gamma_i)\alpha Y) + p_G(1 - \gamma_i)\alpha Y + p_G c_j) = 0 \end{aligned} \quad (19)$$

By simplifying, since:

$$\begin{aligned} (U_i - \bar{U}_i) &= (\gamma_i + v - 1)(\alpha p_k Y + (1 - \alpha) p_R Y + 2 p_G \alpha Y) \\ (U_j - \bar{U}_j) &= (1 - \gamma_i)(\alpha p_k Y + (1 - \alpha) p_R Y) - p_G(c_j - (1 - \gamma_i)\alpha Y) \\ &\quad + p_G(1 - \gamma_i)\alpha Y + p_G c_j \end{aligned} \quad (20)$$

By totally differentiating the first-order condition, we can perform comparative statics in equilibrium and see the effect of the different variables of the model on the equilibrium share of prosumer profits. Starting with the impact of the prosumer's exogenous bargaining capacity, we have that

The comparative statics at equilibrium therefore obviously shows that $\frac{d\gamma_i^*}{d\beta} > 0$, that is, the prosumer share in equilibrium is higher, the higher her exogenous bargaining power. Note as well that $\frac{d\gamma_i^*}{dI} = 0$ and $\frac{d\gamma_i^*}{dr} = 0$, that is, investment cost and the interest rate do not affect the prosumer equilibrium share in the REC game since, under the assumption $U_i'' > 0$, the prosumer finds it convenient to create the REC even if the passive consumer does not participate and nonetheless incurs the investment cost. When this is the case, a change in I or r does not change her gain from the agreement. The situation changes in the Stag Hunt game region ($U_i'' < -pGC_i$) since, in this case, for the prosumer, it is only convenient to create the REC if the passive consumers participate in the venture. In this case, $\frac{d\gamma_i^*}{dI} > 0$ and $\frac{d\gamma_i^*}{dr} > 0$ since a higher investment or interest rate cost reduces the gain from agreement for the prosumer, thereby increasing her bargaining power and reducing her equilibrium share.

Moreover, comparative statics around the total consumption of passive consumers show that $\frac{d\gamma_i^*}{dC_j} = 0$. An increase in the total energy consumed by the passive consumer makes the CER less valuable for him, but at the same time, it reduces the value of the outside option. The two effects exactly offset each other.

Note also that $\frac{d\gamma_i^*}{dY} > 0$ since an increase in the REC output (i.e., a higher productive capacity of prosumers) raises the value of the agreement for the passive consumers without affecting that of the prosumers when we are in the REC game region, in this case, they start and operate the REC even in the absence of passive consumers. In a similar way, $\frac{d\gamma_i^*}{d\alpha_2} > 0$ since when we are in the REC region, the effect of an increase in the accumulation technology (an increase in the green area) raises the value of agreement for the passive consumers, while not for the prosumer since even in the absence of the passive consumer, he/she will create the REC. Hence, technical progress leading to an improvement in the accumulation technology raises the bargaining power of the prosumer.

Consider also that a shock raising the prices of fossil fuels (as in the Russian-Ukrainian war crisis) has a substantial positive impact on the gain from agreement for the passive consumers, while not for the prosumer, thereby raising the bargaining power of the prosumer, hence $\frac{d\gamma_i^*}{dp_G} > 0$.

Comparative statics related to the effect of changes in the self-consumption share ($\frac{d\gamma_i^*}{d\alpha}$), the self-consumption subsidy ($\frac{d\gamma_i^*}{dk}$), and the price of the surplus sale are slightly more complex, but they all show under reasonable conditions that an increase in the self-consumption

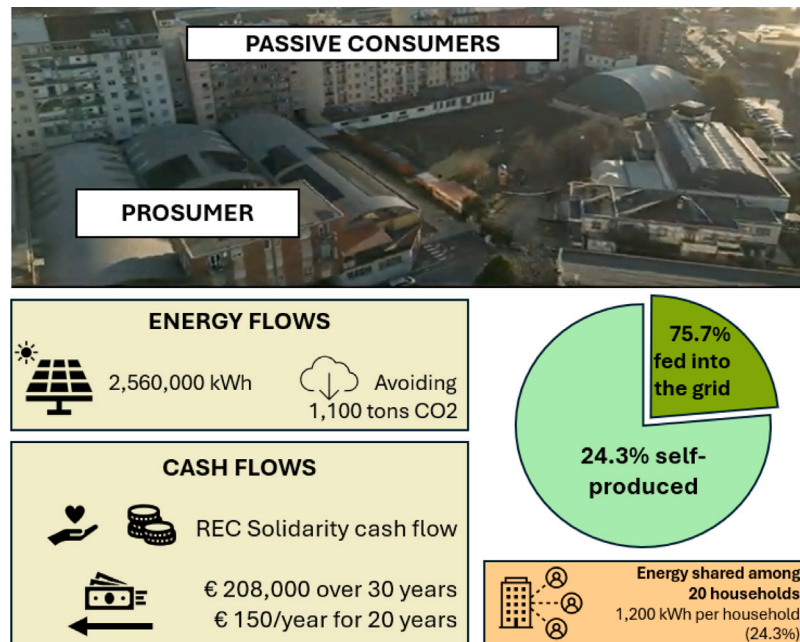


Fig. 4. An overview of the Comunità Solidale di Energia Rinnovabile (CSER) in Baranzate, Milan, Italy.
 Source: Adapted from <https://larotonda.org/it/attivita/sol>.

share and subsidy raises the gain from the agreement, which is conversely decreased by a higher price.

7. A case study

In 2024, the company 3E Environment Energy Economy Ltd started a REC as a single prosumer in Baranzate, Milan, Italy. The REC was called Comunità Solidale di Energia Rinnovabile (CSER) and was established in partnership with La Rotonda, which acted as an energy consumer. The photovoltaic system was installed on the roof of La Rotonda, which offered its buildings for the REC, and had a total minimum peak power of approximately 86–92 kW. Overall, the system today consists of 212 modules, each with a minimum power output of 410 W, guaranteed for 30 years. An electric meter installed at the prosumer's location is used to monitor the energy production, consumption, grid injection or withdrawal, and limit grid injections. The device provides high-precision readings and supports a wide range of current transformers (from 50 to 1000 A), making them compatible with both residential and commercial systems. The meter is equipped with technology that enables monitoring and management of the electrical loads of community members. Fig. 4 shows an overview of the CER.

The REC was initially designed for 20 members, and each member of the REC also had a meter installed. Passive consumers were initially selected through a Web announcement, followed by face-to-face local meetings for all interested participants, which could potentially be part of the REC in the area. Consider that (in Italy and many other countries) it is possible to create a REC only for those that are in the same primary (high voltage) substation, that is, the facility where electricity is transformed from high voltage (transmission) to medium voltage (distribution). Hence, there was a geographically delimited area where potential REC participants could be recruited. The meter monitors all household energy flows, automatically identifies potential energy savings, and optimizes the use of photovoltaic energy. The system optimizes self-consumption within REC by forecasting potential savings and integrating with smart home devices, which can be combined with appliances and machinery for more efficient energy use. Over the past 30 years, the 87 kWp photovoltaic system installed at La Rotonda is

estimated to have generated approximately 2,560,000 kWh of renewable electricity. In terms of environmental benefit, this is equivalent to avoiding 1100 tons of carbon emissions into the atmosphere.

CSER is a Renewable Energy Community whose purpose is to redistribute the revenues generated from incentives for shared energy and/or energy fed into the grid (net of expenses and according to the proportions set by the members' assembly) to needy families who join CSER free of charge. The primary goal of a CSER, therefore, is social. CSER members continue to pay their electricity bills but receive an annual financial contribution from CSER, amounting to over 30 percent of the average annual electricity bill per household.

7.1. The profitability analysis

The profitability analysis of the photovoltaic system, with a minimum peak power of 86–92 kW, installed on the shadow-free area of APS La Rotonda in Baranzate, is limited to the first 30 years of operation (the full warranty period of the photovoltaic modules) and is based on the following factors: (i) the investment is entirely funded through donations (own capital); (ii) the first year production is 91,831 kWh, estimated going up to 78,740 kWh the 30th year, considering the degradation of photovoltaic modules, prudently calculated based on the manufacturer's warranty. Direct self-consumption and subsequent electricity bill savings for REC members (both prosumers and passive consumers) account for 24.29% of the total energy produced. The excess energy fed into the grid amounts to 75.71 percent. The energy shared between the 20 households initially associated with CSER is approximately 1200 kWh per family per year (25.24 percent).

Other crucial ingredients for the profitability analysis are: (i) the fixed energy purchase price for 30 years (€0.2198/kWh); (ii) the fixed energy sales price for 25 years (approximately €0.1099/kWh) for energy produced and fed into the distribution grid. The energy shared between the 20 initially associated users of CSER is 100 percent of the energy fed into the grid. The Italian government incentive for virtual self-consumption over 20 years is €0.13/kWh of shared energy, while the remuneration for shared energy (compensation for avoided transmission costs) by the Italian Regulatory Authority for Energy, Networks and Environment (ARERA) is €0.00884/kWh.

Based on these assumptions, during the 30 years, approximately 1,900,000 kWh of electricity produced by the photovoltaic system will be fed into the grid. The revenues from selling energy to the grid, allocated to a solidarity fund aimed at preventing evictions and addressing housing-related debt situations, are expected to amount to approximately €208,000 over 30 years.

Each household member of the REC is expected to consume, on average, 1200 kWh per year during daylight hours, while the photovoltaic system continues to feed electricity into the grid. The household is also expected to receive an annual financial contribution of approximately €150 for a period of 20 years.

7.2. Lessons learned and links with the model

The case of 3E Environment Energy Economy Ltd. in Baranzate, Milan, embodies several key aspects of the theoretical model developed in this paper, highlighting the strengths and weaknesses of RECs. First, it provides real-world evidence that prosumers often take the lead in initiating RECs, as predicted by our game-theoretic framework. Our model identifies this as a dominant strategy, which, under favorable conditions, creates an equilibrium where passive consumers also find it optimal to join.

Second, the case study aligns with our findings on profit-sharing and bargaining dynamics. The participation of passive consumers in this REC demonstrates that strategic surplus distribution plays a crucial role in ensuring participation. As our model predicts, the bargaining power of passive consumers depends on key factors, including energy pricing, government incentives, and self-consumption subsidies. In this case, the REC was designed to redistribute revenues among participants, including financial contributions to members and a solidarity fund. Note that without the voluntary decision of members to devolve part of the revenues into a solidarity fund, their personal gains from joining the REC would have been much higher. This practice mirrors the theoretical conditions where fair revenue allocation prevents potential conflicts and ensures the long-term stability of the REC.

Finally, the project highlights the policy implications derived from our model. The case demonstrates that self-consumption subsidies and surplus sale pricing play a crucial role in REC viability, supporting our argument that well-calibrated incentives are essential to sustain cooperation. In general, this real-world example demonstrates how crucial factors for the survival of RECs can be accurately analyzed using a game-theoretic approach, highlighting the influence of investment decisions, strategic participation, and revenue-sharing mechanisms on the feasibility of RECs. The insights gained from this case reinforce our policy recommendations, emphasizing the importance of optimizing incentive structures, ensuring fair distribution of surplus, and fostering technological innovation to enhance the effectiveness of RECs in accelerating the ecological transition.

8. Discussion and policy implications

A first straightforward policy implication of the model and of the illustrated case study is that the self-consumption incentive and the surplus sale price are crucial to enhance the creation of RECs and thus cooperation in ecological transition. These two variables are crucial policy choices, since in most countries the regulator sets the surplus sale price and the self-consumption incentive. In our model we outline conditions by which values for these two policy measures determine the switch from the Stag Hunt Game region (where creating a REC is not a dominant strategy for the prosumer) to the original REC game region, where creating a REC is the prosumer dominant strategy and therefore the game has a unique Nash Equilibrium where both prosumers and passive consumers participate to the REC. Entering this region enhances conditions for cooperation since in the Stag Hunt Game there is a dilemma between two Nash equilibria where both parties either create or do not create the REC, and the cooperative solution

of the model in the Stag Hunt game region requires a binding contract with penalties in case of breach where parties commit to create the REC. The straightforward policy advice in this case is to set policy incentives so that we are just in the REC region, thereby maximizing cooperation while minimizing government expenditure.

Consider as well that in our simple theoretical framework, we consider that the different (red, blue, light blue, yellow) areas are fixed, while in reality they can vary according to strategies of REC members that react to the incentive system. More specifically, the system of incentives assumed in our model (and generally adopted in existing RECs worldwide) has several effects. First, it encourages members to minimize the red and maximize the blue area by adjusting their consumption habits, aiming to match consumption as closely as possible with production. Second, it encourages members to expand the light blue area, that is, to adopt new accumulation technologies that can increase the energy produced, accumulated and not purchased from external sources. This outcome of the model is desirable, as it fosters the diffusion and adoption of new accumulation technologies. However, a less desirable outcome of the REC incentive system can also lead to an increase in total energy consumption covered by the participants (the sum of blue, yellow, and light blue areas), which can eventually lead to an overall net increase in energy consumption compared to what happened before the creation of the REC. This adverse effect can be mitigated if the self-consumption incentive is excluded above a maximum threshold of energy consumption (i.e., if the energy consumed by REC members exceeds a certain percentage of what was consumed in the previous year).

Our model has general equilibrium implications that need to be taken into account. A high number of RECs created in a short time interval can create demand pressure, thereby raising investment costs and public expenditure. The government should address the problem by setting a maximum target for the energy that RECs can produce within a given time interval. This target could be related to the EU target mentioned in the introduction (16 percent of renewable energy produced by RECs). We could also reason about the effects of likely progress in accumulation technologies, which will increase the light blue area in the future and, with it, the bargaining power of consumers in RECs. A government decision to extend self-consumption subsidies to the energy produced and accumulated by the REC should further increase the prosumer's bargaining power, but more importantly, enhance the general equilibrium problem discussed above, especially considering the pace of progress in accumulation technologies.

Overall, in the discussion of our model, we find several reasons for moderating the self-consumption subsidy above the lower boundary that brings the model in the original REC equilibrium out of the Stag Hunt Game region. There are general equilibrium and government budget constraint reasons for this, and, in addition, the fact that an increase in the subsidy can lead to excessive bargaining power for prosumers, reducing REC benefits for passive consumers who are likely to belong to lower-income households that cannot afford the initial investment costs. This last consideration implies that there can therefore be distributive reasons for which it could not be advisable or desirable to reduce the bargaining power of passive consumers too much. A final reason suggesting moderation in the self-consumption subsidy is that, as occurred in the past with feed-in tariffs in several countries, governments can decide in the future when realizing they face too high public expenditure, to make a downward revision of the subsidy (see the case of Germany discussed in the literature review). In many cases, such a revision affects not only future RECs but also ongoing RECs as well.

Prosumers and passive consumers can formulate expectations by recognizing the impermanence of current levels of self-consumption subsidies when evaluating expected REC profits. Consider also that the government can create an admission threshold to the REC based on a minimum share of self-consumption to foster aggregation of participants (as it could be in our case study, where the share of

self-consumption is low and can be increased). If this minimum share cannot be achieved by prosumers only, the effect in our bargaining model is that of raising the bargaining power of passive consumers since the rule does not affect the gain from agreement for the passive consumer, while it has a strong effect on the gain from agreement for the prosumer (being equivalent to bringing the model into the Stag Hunt Game region). If low-income households are prevalent among passive consumers, the distributive effect of this choice can be desirable for the government.

Another variant to consider is the presence of switching costs for passive consumers if migration from one model to the other is economically or even only psychologically costly. When this is the case, nonzero switching costs reduce the value of the agreement for passive consumers, while not affecting that of prosumers; thereby weakening prosumer bargaining power and increasing the passive consumer's equilibrium share.

A further issue to be taken into account is the birth of RECs where the investment cost is covered by a government or a private foundation grant (as it is in our case study). In this case, the REC has a special and acknowledged social value (that is, part of the revenues are channeled into a solidarity fund to address energy poverty or other special social purposes). The move does not affect the passive consumer gains from the agreement; however, the same applies to prosumers, who may still find it convenient to create the REC without the participation of passive consumers. However, the social goal of RECs subsidized with a grant can include the participation of passive consumers, and when this occurs, the bargaining power of the latter grows.

Among final policy priorities that could speed the development of RECs are net metering (the possibility of having a direct discount in energy bill when consuming one's production instead of a delayed reimbursement), the reduction time lags in authorizations and the creation of proper information infrastructure such as an electronic register of potential energy power of domestic roofs.

In summary, our model demonstrates that RECs can significantly impact the strategic landscape of ecological investment. By transforming coordination failures into cooperative equilibria, and by aligning surplus-sharing through rational bargaining mechanisms, RECs become viable and stable solutions to the environmental cooperation failure. Our findings emphasize the critical role of subsidy calibration, technological adoption (such as storage), and equitable revenue distribution in determining the success and inclusiveness of RECs. These insights not only validate the theoretical approach but also offer actionable policies to accelerating decentralized energy transitions.

9. Conclusions

The increase in the share of energy produced with renewables is a crucial intermediate target to achieve the goal of net zero emissions needed to tackle the global warming challenge. This intermediate target will be achieved, according to international institutions, by a combination of large plants and renewable energy communities. The role of the latter is deemed essential for generating a decentralized system of energy production and self-consumption, which creates an energy grid that is less congested, more flexible, and more efficient. The development of RECs is equally important in tackling the NIMBY phenomenon and increasing grassroots consensus and participation in the ecological transition.

In our paper we provide a theoretical framework showing how the RECs, with a proper subsidy setting, have the advantage of modifying the structure of the ecological transition game transforming it from a multiplayer Prisoners' Dilemma (where cooperation is difficult to achieve) into a Stag Hunt Game, and eventually an original REC game where the cooperative equilibrium is the unique Nash equilibrium. In developing of our model, we calculate the bargaining equilibrium shares of REC revenues between prosumers and passive consumers, and discuss how various factors influence them. We also demonstrate,

through a case study analysis, that our model captures the crucial variables that determine the benefits to participants, as well as the distinct roles of prosumers and active consumers in creating a REC.

Based on our model, we outline an optimal setting of government policies to make the RECs a vehicle of accelerated ecological transition. A fundamental point is the calibration of the system of incentives above the minimum, which brings the payoff matrix of REC players out of the Stag Hunt Game region into the REC region, where cooperation is more easily achievable and is the unique Nash equilibrium of the game. At the same time, an upward cap on the maximum volume of GW generated by the RECs within a given time interval is fundamental to avoid adverse general equilibrium effects on input markets and the government budget. We also suggest extending the self-consumption incentive to the accumulated energy that is not consumed immediately to stimulate the adoption of improved renewable energy storage technologies.

With our model, we intended to outline and discuss the main issues at stake in the birth of a REC. A limitation of our model is its partial equilibrium analysis, despite our discussion considering general equilibrium effects. Further research formally incorporating such effects in the theoretical framework is welcome.

CRedit authorship contribution statement

Leonardo Becchetti: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Francesco Salustri:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Leonardo Becchetti reports financial support was provided by European Union. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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